# Stress Worksheet

When a force is applied to a part of a structure, it will experience **STRESS**. If the level of force which is applied to the member is too high then the stress which is created may cause the part to bend, buckle or worst case, fail and break.

The stress can vary by the force put on it, but also by the size of the structure. For example; if the same load is applied to column A and column B which are of the same length, then Column A will suffer **more** from stress and will be more likely to fail. The reason why is because Column A has a far smaller cross-sectional area.

Where Force is measured in Newtons (N) and the Area is the cross-sectional area measured in mm2. Stress therefore is measured in $Nmm^{-2}$ (the unit measurement can also be given as $N/mm^2$). This means the amount of force acting on a structure is Newtons '**per**' square mm). Stress is denoted by the Greek letter Sigma ( $\sigma$ ). It is calculated using the formula:

**Stress = Force / Area**

**In your jotter, copy the follow questions and answer.**

1. A rectangular bar of 180mm x 120mm cross section is subjected to a tensile load of 500N. Calculate the stress in the bar.

   Force = 500N

   Area = 180 × 120mm

2. A wire 4.0mm in diameter is subjected to a force of 300N. Calculate the stress of the wire.

   Force = 300N

3. The stress in a steel wire supporting a load of 8.0kN should not exceed 200N/mm$^2$. Calculate the minimum diameter of wire required to support the load.

   Force = 8.0kN = 8000N

   $\sigma = 200 Nmm^{-2}$

*Although we should probably round up for safety reasons

4. A column of cross section 0.73m$^2$ is required to act as a roof support. The maximum allowable working stress in the column is 4.0 MNm$^{-2}$. Calculate the maximum compressive load acting in the column.

   Area = 0.73m$^2$

   $\sigma = 4.0 MNm^{-2} = 4{,}000{,}000 Nm^{-2}$

5. The figure shows an oil platform used in the North Sea. The platform is designed so that each of the three legs always carries an equal share of the **120 MN** load. The legs are **15.0m** in diameter and are made of steel reinforced concrete. Calculate the stress that is applied on **each leg.**

   Force = 120MN = 120,000,000N

   Force per leg = 120,000,000 / 3 = 40,000,000N